

Qylith: Quantum-Secure AI-Native Layer 1 Blockchain

Whitepaper v1.0

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"We are not preparing for a future threat—we are standing at the threshold of the quantum era from the genesis block."

"We are not 'supporting' AI Agents. We are building the digital foundation for an AI-native civilization."

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Executive Summary

The Convergence of Two Existential Threats

We stand at a critical inflection point in digital infrastructure. Two transformative forces—quantum computing and artificial intelligence—are converging, and the existing internet landscape, including virtually all blockchain networks, is woefully unprepared.

The Quantum Threat is no longer theoretical. In March 2026, Google’s Quantum AI team published a landmark paper demonstrating that a fault-tolerant quantum computer with 500,000 physical qubits could break ECDSA signatures within acceptable timeframes. This means the cryptographic foundations of Ethereum, Solana, Bitcoin, and virtually every other blockchain face existential risk between 2027 and 2030.

The AI Agent Revolution is accelerating. AI agents are evolving from experimental concepts into autonomous economic actors capable of independent decision-making, collaborative work, and asset management. Yet no existing blockchain provides a native execution environment for AI agents—they are merely “borrowed” tools, like running modern software on DOS.

Introducing Qylith

Qylith is the answer. We are not a blockchain that has been retrofitted with quantum resistance or awkwardly bolted with AI features. Qylith is a Layer 1 infrastructure designed from the ground up with post-quantum cryptography (PQC) and an AI-native execution environment as core architectural axioms from the genesis block.

Key Differentiators:

Feature	Traditional Blockchains	Qylith
Cryptographic Foundation	ECDSA/Ed25519 (quantum-vulnerable)	FALCON-1024 + ML-DSA-87 (NIST PQC)
AI Agent Support	Application-layer afterthought	Protocol-native execution environment
Cryptographic Migration	Hard forks or patches required	Native from genesis, no migration needed
AI Inference Verification	None	STARK-based on-chain verification
Agent Identity	Wallet proxy for humans	Native on-chain Agent identity system

Our Commitment:

- **Quantum Security:** We implement NIST-standardized post-quantum cryptography (FALCON-1024 for signatures, ML-KEM-768 for key encapsulation) from day one
- **AI-Native Architecture:** Our AI Agent Execution Module (AEM) provides native identity, reputation, inference verification, and autonomous execution capabilities
- **Crypto-Agility:** Built-in upgradability ensures we can adapt to future cryptographic advances without disrupting the network
- **Performance:** Target of 10,000+ TPS through parallel execution engine

Invitation to Builders

This whitepaper presents our technical architecture, token economics, application ecosystem, and development roadmap. We invite global builders, cryptographers, AI researchers, and visionaries to join us in this foundational infrastructure revolution that will shape the next decade of the internet.

Chapter 1: Problem Statement

Why Qylith Must Exist

1.1 The Quantum Threat: Not “If,” But “When”

Before 2026, the quantum threat was considered a distant future concern. After March 2026, this perception was permanently shattered.

Google Quantum AI’s landmark paper, “*Cryptographically Relevant Quantum Computing: Timeline and Implications*,” provides precise timeline analysis:

Quantum Threat Timeline

gantt		
title Quantum Computing Development & Blockchain Impact		
dateFormat YYYY-QQ		
axisFormat %Y		
section Quantum Milestones		
1,000 Physical Qubits (Theoretical Threat)	:	2026-Q2, 2026-Q4
50,000 Physical Qubits (ECDSA Breakable)	:	2027-Q1, 2029-Q2
1M+ Physical Qubits (All Non-PQC Vulnerable)	:	2029-Q3, 2032-Q4
section Action Window		
Build Qylith Now	:	active, 2026-Q2, 2028-Q2
Last Chance for Migration	:	2026-Q2, 2029-Q4

Quantum Threat Timeline - Technical Specification

Quantum Computing Milestone	Estimated Timeline	Impact on Blockchains
1,000 Physical Qubits	2026-2027	Theoretical threat emerges; no practical attack yet
50,000 Physical Qubits	2027-2029	ECDSA signatures breakable ; private keys at risk
100,000+ Physical Qubits	2029-2032	All non-PQC cryptography compromised

Critical Conclusion: A fault-tolerant quantum computer with just 500,000 physical qubits can break ECDSA signatures. This means:

- Attackers can forge signatures and steal assets from any address
- Historical transactions can be retroactively altered (“harvest now, decrypt later” attacks)
- Cross-chain bridge trust assumptions are completely invalidated
- The entire DeFi ecosystem becomes vulnerable to unprecedented attacks

Current Industry Status: Ethereum, Solana, Bitcoin, and all major blockchains rely on ECDSA secp256k1 or Ed25519 for transaction signatures. These algorithms are as secure as paper against quantum computers.

The Urgent Action Window

The window to build a quantum-resistant chain is only 2-3 years. Yet existing “quantum-resistant” solutions have fatal flaws:

1. **Temporary Patches:** Adding PQC layers to existing chains introduces complex

- technical debt and compatibility issues
- 2. **Fork-Dependent Transitions:** Require hard forks, facing community fragmentation risks
- 3. **Non-Native Design:** PQC is an add-on rather than a core architectural component

Qylith's Answer: Native integration of NIST-standardized post-quantum cryptography from the genesis block. No migration. No forks. No compromise.

1.2 AI Agent Explosion: The Lack of Native On-Chain Execution Environment

2025-2026 marks the transition of AI agents from laboratory experiments to market-ready products. They now execute transactions autonomously, manage assets, and collaborate. However, current blockchain architectures were designed for humans—AI agents merely “borrow” these tools, like running modern software on DOS.

The AI Agent Dilemma

THE FOUR AGENT DILEMMAS
PROBLEM 1: IDENTITY DILEMMA • Humans prove identity with private keys • How do AI Agents obtain verifiable identity? • Current "wallet" solutions are proxies for human identity
PROBLEM 2: EXECUTION DILEMMA • Smart contracts execute deterministically • AI inference is inherently non-deterministic • How do we verify correctness of on-chain AI inference?
PROBLEM 3: COLLABORATION DILEMMA • How do multiple AI Agents trust each other? • How do we coordinate resource allocation? • How do we establish agent-level reputation systems?
PROBLEM 4: RESOURCE DILEMMA • AI inference computation is expensive • Traditional gas models cannot meter AI operations • A dedicated AI execution environment is required

Current Industry Status: The “AI + Blockchain” integration remains superficial:

- AI-generated NFTs (creation layer)
- AI analysis of on-chain data (tool layer)
- AI-driven investment strategies (application layer)

These represent AI’s “external empowerment” of blockchain, not blockchain’s “native support” for AI.

Qylith's Answer: Protocol-level native integration of AI Agent execution environment. The AEM (AI Agent Execution Module) is not an application—it is infrastructure. Just as Ethereum’s EVM is the foundation for smart contracts, AEM is the womb for AI agents.

1.3 First Principles: Why This Blockchain Must Exist

Let us return to the essence of the problem:

What is blockchain's value proposition? > A trustless, censorship-resistant, globally accessible system for value storage and transfer.

What is the foundation of this value proposition? > Cryptographic security.

If cryptography is broken, what remains of blockchain? > Nothing.

What is AI Agent's value proposition? > Autonomous, sustainable, collaborative intelligent execution.

What is the foundation of this value proposition? > Native digital identity, verifiable execution, and reliable collaboration mechanisms.

If AI Agents must work with human tools under human rules, can they ever be truly "autonomous"? > No.

Conclusion:

1. Quantum resistance is not optional—it is a survival imperative
2. AI-native capability is not a bonus—it is the entry ticket to the next era
3. Existing chains cannot satisfy both conditions because they are retrofit solutions
4. The only correct approach: design from scratch for the new era

Qylith is the First and Only: A Layer 1 blockchain designed from inception with quantum resistance and AI-native capability as core axioms, not afterthoughts.

Chapter 2: Vision & Mission

2.1 Vision

To build the world's first quantum-secure, AI-native Layer 1 blockchain infrastructure, serving as the value layer and execution layer for digital civilization over the next decade.

We believe that by 2030:

- Over 1 billion AI agents will be active in the digital world
- Quantum computing threats will become a core consideration for all financial systems
- Quantum-resistant, AI-native infrastructure will become as fundamental as TCP/IP is today

Qylith aspires to be this future's protocol layer.

2.2 Mission

1. **Security Mission:** Provide a quantum-safe harbor for global users before the quantum era arrives
2. **Empowerment Mission:** Deliver native, seamless, verifiable execution environment for AI agents
3. **Ecosystem Mission:** Build an open, sustainable community of developers and users
4. **Evolution Mission:** Ensure long-term technological leadership through crypto-agility

2.3 Core Values

- **Security First:** No compromise on security for any design decision
- **Open Source & Transparency:** All core protocol code open-sourced; open governance
- **Community Driven:** Major decisions through on-chain governance voting
- **Long-Termism:** No sacrifice of long-term vision for short-term gains

Chapter 3: Technical Architecture Overview

3.1 Architectural Philosophy

Qylith’s technical architecture adheres to these core principles:

1. **Cryptography-Native:** Post-quantum cryptography is not a layer or module—it is embedded as the foundation of every layer
2. **AI First-Class Citizen:** AI agents are natively supported at the protocol layer, not as application-layer add-ons
3. **Modular Design:** Layers evolve independently, communicating through standard interfaces
4. **Performance Scalable:** Parallel execution and state management optimization enable high TPS

3.2 Layered Architecture

Application Layer (DeFi, AI Agent Market, RWA, Social, etc.)
Execution Layer WASM Runtime + AI Agent Execution Module (AEM) Parallel Execution Engine (PEE)
Consensus Layer Nominated Proof of Stake (NPoS)
Cryptographic Layer FALCON-1024 ML-DSA-87 ML-KEM-768 Hybrid Signature Schemes
Network Layer libp2p + PQC-Encrypted P2P Protocol
Data Layer MMR + State Expiry + Quantum-Secure Commitment

Chapter 4: Cryptographic Layer

The Quantum-Era Security Foundation

4.1 Why NIST Post-Quantum Cryptography Standards

In 2024, NIST officially published post-quantum cryptography standards—the crystallization of 30 years of global cryptographic research. Unlike academic algorithms, NIST standards have undergone:

- Public analysis by cryptographers worldwide
- Multiple rounds of selection and optimization

- Optimal balance of security and performance

Qylith comprehensively adopts NIST standards, ensuring:

- **Legal Compliance:** Prerequisite for government procurement and enterprise adoption
- **Long-Term Security:** Extensive validation by academic and industrial communities
- **Interoperability:** Future integration with other PQC systems

NIST PQC Standards Adopted by Qylith:

Standard	Algorithm Type	Use Case in Qylith	Status
FIPS 206	FALCON-1024	Transaction signatures	Primary signer
FIPS 204	ML-DSA-87	Hybrid signatures, fallback	Secondary signer
FIPS 203	ML-KEM-768	Key encapsulation	Network encryption

Reference: [NIST Post-Quantum Cryptography Standards](#)

4.2 Transaction Signing: FALCON-1024

Algorithm: Fast Fourier Lattice-based Compact Signatures over NTRU **Parameter:** FALCON-1024 (Category 5, 128-bit security level)

Why FALCON-1024?

- **Smallest Signatures:** 1,280 bytes—47% smaller than ML-DSA-87
- **Lightning-Fast Verification:** 0.15ms—6-7x faster than ML-DSA-87
- **NIST Level 5:** Highest security classification
- **Robust Theoretical Foundation:** Based on NTRU lattice problems

ML-DSA-87 Retained as Fallback: Provides ECDSA+ML-DSA hybrid signature compatibility during transition period

Applications in Qylith:

- User transaction signatures
- Governance vote signatures
- Cross-chain message signatures
- AI Agent identity binding

Parameter Specifications:

FALCON-1024 Parameters:

Public Key Size:	1,793 bytes
Secret Key Size:	~2,400 bytes
Signature Size:	1,280 bytes
Security Level:	NIST Level 5 (128-bit)
Hard Problem:	NTRU Shortest Vector Problem
Sign/Verify Speed:	~0.1ms / ~0.15ms

4.3 Key Encapsulation: ML-KEM-768

Algorithm: Module-Lattice Key Encapsulation Mechanism **Parameter:** ML-KEM-768

Selection Rationale:

- Same lattice family as ML-DSA, simplifying implementation

- Excellent key encapsulation speed, suitable for high-frequency network communication
- High NIST standardization maturity

Applications in Qylith:

- Inter-node TLS-like encrypted channels
- Cross-chain bridge key negotiation
- Secret sharing and threshold signatures

Parameter Specifications:

ML-KEM-768 Parameters:

Public Key Size:	1,184 bytes
Ciphertext Size:	1,088 bytes
Shared Secret:	32 bytes
Security Level:	NIST Level 3 (128-bit)
Encapsulation Speed:	~0.1ms
Decapsulation Speed:	~0.2ms

4.4 Transition Strategy: Hybrid Signatures

Challenge: Users may already have ECDSA keypairs. How do we enable smooth migration?

Solution: Hybrid Signature Mechanism

How It Works:

Hybrid Signature = ECDSA Signature ⊗ ML-DSA-87 Signature

Verification: Both signatures must verify successfully = Valid signature

Advantages:

1. Compatible with existing wallets (MetaMask, etc.)
2. Double protection (if either algorithm is broken, the other remains secure)
3. Smooth transition (no hard fork required)

Transition Timeline:

Phase	Timeline	Supported Signatures	Notes
Phase 1	Genesis Block	ML-DSA-87 only	Native PQC-only launch
Phase 2	6 months post-mainnet	ECDSA + ML-DSA hybrid	Backward compatibility enabled
Phase 3	18 months post-mainnet	ML-DSA-87 only	ECDSA deprecated

4.5 Crypto-Agility: Future-Proof Upgradability

Challenge: Stronger PQC algorithms may emerge in the future, or weaknesses may be discovered in current algorithms.

Solution: Crypto-Agility Architecture

Design Principles:

1. **Algorithm Abstraction Layer:** Protocol layer does not hard-code specific algorithms

2. **Versioned Signature Format:** Signatures include algorithm identifier
3. **On-Chain Upgrade Mechanism:** Cryptographic parameters upgradeable through governance voting
4. **Backward Compatibility:** Old signature formats continue to function

Upgrade Pathway:

Current: ML-DSA-87 + ML-KEM-768
↓ (via governance vote)
Future: Potential upgrade to more efficient lattice algorithms
↓
Even Further: Algorithm rotation based on emerging standards

Reference: [Crypto-Agility Best Practices](#)

Chapter 5: Consensus Layer

Nominated Proof of Stake (NPoS)

5.1 Why NPoS

Qylith selects NPoS over traditional PoS or PoW based on the following considerations:

1. **Security:** NPoS's validator election mechanism provides stronger attack resistance than traditional PoS
 2. **Decentralization:** The nominator mechanism allows small QYL holders to participate in security maintenance
 3. **Energy Efficiency:** Pure proof-of-stake, no physical computing power required
 4. **Quantum Resistance Foundation:** PoW's SHA-256 is accelerated by quantum computers (Grover's algorithm); NPoS relies on signature security and is unaffected by quantum attacks
-

5.2 NPoS Mechanism Details

Role Definitions:

- **Validator:** Runs a full node, participates in block production and finality confirmation
- **Nominator:** Stakes QYL to support validators, shares rewards and risks

Election Mechanism:

- Each era (6 hours), the validator set is re-elected
- Phragmén's election algorithm based on available stake + historical performance
- Ensures fair distribution of the validator set

Staking Parameters:

Staking Parameters:

Minimum Validator Stake:	100,000 QYL
Minimum Nominator Stake:	1,000 QYL
Maximum Validators:	1,000
Max Nominators per Validator:	16
Max Validators per Nominator:	256

5.3 Block Production & Finality

Timing Parameters:

- **Slot:** 6 seconds (one block production cycle)
- **Epoch:** 300 slots (~30 minutes)
- **Era:** 12 epochs (~6 hours)

Finality Guarantee:

- Blocks require 2/3+ validator signature confirmation
- Confirmation delay: 12-18 seconds (2-3 blocks)
- Finality gadget (GRANDPA-style) ensures irreversibility

Performance Metrics:

Performance Specifications:

Target TPS:	10,000+ (via Parallel Execution Engine)
Actual Block Time:	6 seconds
Finality Confirmation:	12-18 seconds
Maximum Validators:	1,000+
Annual Staking Yield:	~12-15%

Chapter 6: Execution Layer

AI-Native Smart Contracts & Agent Environment

6.1 Why AI-Native Execution Layer

Problems with Ethereum EVM:

- Designed for deterministic computation
- Cannot natively support AI inference
- No Agent identity or reputation mechanism

AI Agent Requirements:

- Deterministic + Non-deterministic (inference) hybrid execution
- On-chain identity authentication
- Verifiable inference proofs
- Autonomous transaction execution

Qylith's Solution: At the execution layer, simultaneously support:

1. **Deterministic Execution:** WASM smart contracts (EVM-compatible)
 2. **AI-Native Execution:** AI Agent Execution Module (AEM)
-

6.2 WASM Smart Contract Runtime

Why WASM?

- **Performance:** Near-native execution speed
- **Cross-Platform:** Runs in browsers, servers, embedded systems
- **Ecosystem:** Supports Rust, C++, Go, and other languages
- **Determinism:** Can construct deterministic execution environments

Supported Contract Languages:

- **ink!:** Smart contracts in Rust (mature in Polkadot ecosystem)
- **Solidity:** Supported via transpilation or compatibility layer

- **Move:** Potentially introduced in the future

Execution Model:

- Account model (similar to Ethereum)
- Gas metering and execution fees
- Cross-contract calls (via call trees)
- Parallel execution optimization (see 6.4)

6.3 AI Agent Execution Module (AEM)

AEM is Qylith's core innovation—this is not an application, but a protocol-level native module.

AEM's Core Functions:

6.3.1 Agent Identity & Reputation System

On-Chain Agent Identity:

```
// Agent Identity Structure
struct AgentIdentity {
    public_key: ML-DSA-87 PublicKey, // Agent's signing public key
    registration_slot: SlotNumber,   // Registration block
    attestation: Vec<Attestation>,   // Third-party attestations
    reputation_score: u128,           // Reputation score (computed on-chain)
}
```

Reputation System:

- On-chain reputation scores based on historical behavior
- Composable trust graphs (Web of Trust)
- Cross-Agent credit sharing

6.3.2 On-Chain Inference Verification

Challenge: AI inference is inherently non-deterministic. How do we verify it on a deterministic chain?

Solution: STARK Proof Verification

Verification Flow:

1. AI Agent performs inference off-chain
2. Generates STARK proof (proving inference correctness)
3. On-chain verifies STARK proof (without re-executing inference)
4. Verification passed → Results trusted

Advantages:

- No need to run AI models on-chain (prohibitively expensive)
- Verification cost far lower than execution cost
- Trustworthiness mathematically provable

6.3.3 Autonomous Transaction Execution

Authorization Mechanism:

```
// Permission levels for user authorization to Agent
enum AgentPermission {
    ReadOnly,           // Read-only (view balances, etc.)
    TradeOnly,          // Trading only (specific tokens)
```

```
    Unilateral,    // Agent decides autonomously
    Managed,       // Managed (requires user confirmation)
}
```

Security Measures:

- Explicit permission level authorization
- Transaction limit controls
- Time window restrictions
- Emergency freeze mechanism

6.3.4 Multi-Agent Orchestration

Coordination Protocol:

- Agents communicate directly (on-chain messages)
- Support for atomic multi-hop transactions
- Sharded execution for collaborative tasks
- Dispute resolution mechanism

Use Cases:

- DeFi strategies: Multiple agents collaboratively execute complex strategies
- Identity verification: Multi-Agent cross-verification
- Service marketplace: Task delegation between agents

6.4 Parallel Execution Engine (PEE)

Inspiration: MegaETH's parallel EVM approach

Challenge: Traditional blockchains execute transactions serially, leading to low efficiency

Solution: Optimistic Concurrency Control + Dependency Graph Analysis

Execution Flow:

Execution Pipeline:

1. Block producer packages transactions
2. Analyze state dependencies (read/write sets)
3. Construct dependency graph
4. Execute independent transactions in parallel
5. Conflict detection and rollback
6. Commit final state

Performance Improvement:

- Theoretical TPS improvement: 10-50x (depending on transaction dependencies)
- Actual target: 10,000+ TPS

Chapter 7: Data Layer

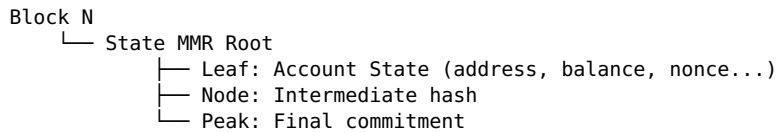
Quantum-Secure State Management

7.1 Merkle Mountain Range (MMR) State Storage

Why MMR?

- Supports incremental appends (append-only)
- Efficient inclusive proofs
- Range query friendly
- State pruning friendly

Structure:



7.2 State Expiry Mechanism

Challenge: Blockchain state grows infinitely, increasing node storage costs

Solution: Epoch-based State Expiry

Design:

State Management:

Active State (Recent 2 epochs):

- Fully readable and writable

Historical State (>2 epochs ago):

- Only MMR root + state proof retained

Recovery Mechanism:

- State can be restored by providing proofs

Benefits:

- Reduces state storage requirements by ~70%
 - Encourages archival nodes (providing proof services)
 - Maintains historical verifiability
-

7.3 Quantum-Secure State Commitment

Challenge: Merkle trees based on SHA-256 have reduced security against quantum computers

Solution: Use Poseidon hash (adopted by zkSync and others)

Poseidon Advantages:

- Designed for zero-knowledge proofs
 - Highly efficient in zkSNARK/STARK systems
 - Superior resistance to quantum attacks compared to traditional hashes
-

Chapter 8: Network Layer

Quantum-Secure P2P Communication

8.1 libp2p + ML-KEM

Why libp2p?

- Production-grade P2P networking library (used by IPFS, Polkadot, etc.)
- Modular design, easy to extend
- NAT traversal, connection management out-of-the-box

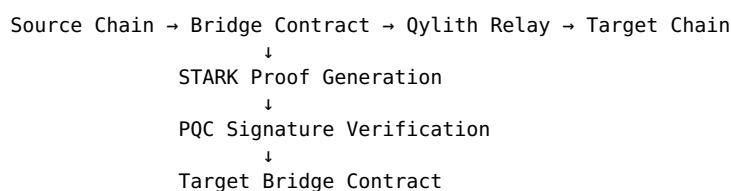
PQC Enhancements:

- Node handshakes use ML-KEM-768 for key negotiation
- Communication content encrypted with symmetric keys (AES-256-GCM)
- Periodic key rotation (every epoch)

8.2 Quantum-Secure Cross-Chain Bridge

Architecture:

Bridge Architecture:



Security Guarantees:

- Cross-chain messages compressed via STARK proofs
- Proof verification completed on Qylith chain
- Relayers require no trust assumptions (trustless)

8.3 Cross-Chain Interoperability

Design Reference: LayerZero, Wormhole multi-chain message passing architecture

Message Passing Flow:

1. Source Chain: User initiates cross-chain transaction
2. Relayer: Collects on-chain events, generates proofs
3. Qylith: Verifies STARK proof + PQC signature
4. Target Chain: Executes corresponding operation

Chapter 9: Token Economics

9.1 QYL Token

Token Information:

Token Specifications:

Name:	Qylith Token
Symbol:	QYL
Standard:	QRT-20 (Qylith native token standard)
Total Supply:	1,000,000,000 (1 billion)
Decimals:	18

9.2 Allocation

Category	Percentage	Quantity	Vesting Schedule
Community & Ecosystem Fund	40%	400,000,000	Gradual release
Ecosystem Development Fund	25%	250,000,000	4-year linear vesting
Investors	20%	200,000,000	1-year lock + 2-year release
Core Team	15%	150,000,000	2-year lock + 3-year release

9.3 Staking Economics

Staking Reward Model:

Annual Percentage Yield (APY): 12-15%

Reward Sources:

- Block rewards (inflation)
- Transaction fees
- AI Agent execution fees

Inflation Model:

- Initial inflation rate: 5%
- Gradually reduce to 2% (long-term equilibrium target)

9.4 AI Agent Gas Model

Innovative Dual-Layer Gas System:

Layer 1: Base Gas (Traditional Smart Contract Execution)

- Similar to Ethereum Gas model
- Priced by opcode complexity

Layer 2: AI Execution Gas (Agent-Specific Operations)

- STARK Verification Gas: Priced by proof size
- Agent Registration Gas: One-time fee
- Reputation Update Gas: On-chain computation fees

Fee Examples:

Fee Schedule:

Standard Transfer:	21,000 Gas
Simple Contract Call:	50,000-200,000 Gas
STARK Verification:	500,000 Gas (one-time verification, reusable)
Agent Registration:	100 QYL (network governance setting)

Chapter 10: Application Ecosystem

Qylith will support the following native application scenarios:

10.1 DeFi Protocols

- **Quantum-Secure AMM:** Quantum-resistant decentralized exchanges
- **Lending Protocols:** AI-driven risk assessment and interest rate pricing
- **Derivatives:** On-chain perpetual contracts, options

10.2 AI Agent Marketplace

- Agent service trading platform
- AI inference compute marketplace

- Agent skill certification and licensing

10.3 Quantum-Secure Cross-Chain Bridge

- Bridges to Ethereum, Solana, Bitcoin
- Asset bridging + message bridging
- Quantum-resistant cross-chain verification

10.4 AI-Driven Insurance Protocol

- Decentralized insurance marketplace
- AI-powered claim verification
- Dynamic premium pricing

10.5 Privacy Transactions

- Zero-knowledge proof-driven private transactions
- Quantum-resistant privacy protection
- Compliance-friendly privacy solutions

10.6 Decentralized AI Training

- Distributed AI model training
- Data privacy protection
- Model copyright and revenue distribution

10.7 Quantum-Secure DID

- Quantum-resistant Decentralized Identity
- AI Agent identity authentication
- Decentralized reputation system

10.8 RWA Tokenization

- Tokenization of real estate, art, commodities
- Long-term asset security
- Institutional-grade custody solutions

Chapter 11: Governance

11.1 On-Chain Governance Mechanism

Governance Participants:

- **Council:** Elected by QYL holders, responsible for day-to-day decisions
- **Technical Committee:** Professional decisions, rapid response
- **General Voting:** Major decisions require all token holder voting

Governance Scope:

- Protocol upgrades
- Cryptographic parameter adjustments
- Treasury fund usage
- Ecosystem fund allocation

11.2 Upgrade Mechanism

Runtime Upgrades:

- Approved through governance voting
- No hard forks required
- Atomic execution
- Rollback mechanism (emergency recovery)

Chapter 12: Roadmap

Phase 1: Genesis (2026 Q3 - Q4)

Milestone	Status
Whitepaper Publication	Complete
Testnet Alpha Launch	In Progress
Developer Documentation & SDK	In Progress
Community Building & Grants Program	In Progress

Phase 2: Exodus (2027 Q1 - Q2)

Milestone	Status
Mainnet Launch	☐ Planned
Core DeFi Applications	☐ Planned
AI Agent SDK Release	☐ Planned
Staking功能开放	☐ Planned

Phase 3: Singularity (2027 Q3 - Q4)

Milestone	Status
Cross-Chain Bridges (ETH, SOL, BTC)	☐ Planned
AI Agent Marketplace Launch	☐ Planned
Ecosystem Project Accelerator	☐ Planned
Governance Handover to Community	☐ Planned

Phase 4: Evolution (2028+)

Milestone	Status
Enterprise Adoption	☐ Planned
Mature AI Agent Ecosystem	☐ Planned
Continuous Protocol Evolution	☐ Planned
Layer 2 Scaling Solutions Exploration	☐ Planned

Chapter 13: Competitive Landscape

13.1 Competitive Analysis

Qylith enters a landscape with several quantum-resistant blockchain projects. Below is our analysis of key competitors:

Feature	Qylith	Quantus Network	QoreChain	Naoris Protocol
Cryptographic Base	FALCON-1024 + ML-DSA-	Lattice-based (custom)	Hash-based + ECDSA + post-Lattice hybrid quantum overlay	

	87 (NIST)			
AI-Native Support	Native AEM	None	Limited SDK	None
Genesis PQC	Yes	No (migration)	No (hybrid)	No (overlay)
Consensus	NPoS	PoS	DPoS	PoH + PoS
Target TPS	10,000+	3,000	5,000	1,000
AI Inference Verification	STARK-based	None	ZK-SNARK	None
Agent Identity System	Native on-chain	None	None	None
Crypto-Agility	Full	Partial	Partial	None

13.2 Key Differentiators

vs. Quantus Network:

- Quantus uses custom lattice-based algorithms; Qylith uses NIST-standardized algorithms for legal compliance and interoperability
- Quantus requires migration from existing chains; Qylith is native from genesis
- Quantus has no AI-native capabilities; Qylith has protocol-level AEM

vs. QoreChain:

- QoreChain uses a hash-based + lattice hybrid approach; Qylith uses a unified NIST standard portfolio
- QoreChain provides limited AI SDK; Qylith delivers native agent execution environment
- QoreChain's hybrid model still carries ECDSA vulnerabilities during transition

vs. Naoris Protocol:

- Naoris adds post-quantum overlay to existing chains (not a new chain); Qylith builds quantum security into the base layer
- Naoris focuses on cybersecurity mesh; Qylith focuses on AI-native infrastructure
- Naoris does not address AI agent execution needs

13.3 Our Positioning

Qylith occupies a unique position as the **only** Layer 1 blockchain that combines:

1. **Genesis-block quantum security** (not migration, not overlay)
2. **Native AI agent execution environment** (not SDK, not application layer)
3. **NIST-standardized cryptography** (for compliance and interoperability)
4. **Full crypto-agility architecture** (for future-proof upgrades)

Chapter 14: Frequently Asked Questions (FAQ)

Technical Questions

Q1: Why did Qylith choose FALCON-1024 over ML-DSA-87 for primary signing?

A: FALCON-1024 offers significant advantages for blockchain applications:

- **Smaller signatures:** 1,280 bytes vs. 2,420 bytes (47% reduction) reduces on-chain storage and transmission costs

- **Faster verification:** ~0.15ms vs. ~1ms makes high-TPS applications feasible
- **Higher security level:** NIST Level 5 vs. Level 3 provides stronger security guarantees
- **Mature implementation:** FALCON's NTRU foundation has been extensively cryptanalyzed

We retain ML-DSA-87 as a fallback and for hybrid signature compatibility during transition.

Q2: How does on-chain STARK verification work, and why is it cost-effective?

A: The key insight is the asymmetry between proof generation and verification:

Off-Chain (Agent):

AI Inference + STARK Proof Generation

Cost: High (one-time, amortized across uses)

On-Chain (Qylith):

STARK Verification

Cost: Low (~500,000 Gas flat fee)

Verification time: Milliseconds

One STARK proof can be verified multiple times by different parties at minimal cost. This transforms expensive AI inference into a verifiable, reusable artifact.

Q3: What happens if a stronger quantum computer than expected emerges?

A: Qylith's crypto-agility architecture addresses this:

1. **Algorithm abstraction layer:** The protocol does not depend on specific algorithms
2. **Versioned signatures:** All signatures include algorithm identifiers
3. **On-chain governance upgrade:** Cryptographic parameters can be upgraded via governance vote without hard forks
4. **Hybrid signatures during transition:** The transition period uses hybrid signatures (ECDSA + ML-DSA) providing double protection

In the worst case, we can upgrade to more conservative parameters or alternative algorithms (e.g., hash-based signatures like SPHINCS+) within our existing governance framework.

Q4: How does Qylith's parallel execution engine handle smart contract conflicts?

A: We use optimistic concurrency control:

1. Read-Write Set Analysis
Each transaction declares which state it will read/write
2. Dependency Graph Construction
Transactions with overlapping state are linked
3. Parallel Execution
Independent transactions execute simultaneously
4. Conflict Detection
If execution results conflict, rollback and retry
5. Sequential Fallback
Conflicting transactions execute serially

In practice, most DeFi transactions have minimal overlap, enabling 10-50x throughput improvement.

Q5: Can existing Ethereum/Solana dApps migrate to Qylith?

A: Yes, through multiple paths:

- **ink! contracts:** Direct migration for Rust-based contracts
 - **Solidity compatibility layer:** For Solidity contracts (with some limitations)
 - **Cross-chain bridges:** Asset and data bridging without code migration
 - **AI Agent wrapping:** Existing dApps can leverage AEM by deploying AI agents that interact with existing contracts
-

Investment & Token Questions

Q6: What is the total supply of QYL and will it be inflationary?

A: Total supply is 1 billion QYL with an initial inflation rate of 5%, gradually decreasing to 2% as the network matures. This follows a similar model to Polkadot's, balancing validator incentives with token value preservation.

Q7: When will QYL be tradeable?

A: Token listing details will be announced closer to mainnet launch. Community and ecosystem allocations are locked according to vesting schedules to protect long-term value.

Q8: What is the minimum stake to participate in network security?

A: As a nominator, minimum stake is 1,000 QYL. This allows broad participation while ensuring validator minimums (100,000 QYL) maintain network security.

Project & Ecosystem Questions

Q9: How does Qylith compare to other quantum-resistant blockchains like Quantus or QoreChain?

A: Please refer to Chapter 13 (Competitive Landscape) for detailed comparison. Key differentiator: Qylith is the only chain with quantum security from the genesis block AND native AI agent support at the protocol level.

Q10: What is the timeline for AI Agent features?

A: AI Agent features are planned for Phase 2 (Exodus) with mainnet launch:

- Phase 1 (Genesis): Foundation, testnet, developer tools
 - Phase 2 (Exodus): AEM core features, AI Agent SDK, basic agent marketplace
 - Phase 3 (Singularity): Full marketplace, multi-agent orchestration, enterprise features
-

Q11: How can developers get involved early?

A: We welcome developer participation:

- **Testnet participation:** Join our testnet alpha program

- **Developer grants:** Apply for Qylith grants for innovative projects
 - **SDK feedback:** Help shape our AI Agent SDK through early access
 - **Community:** Join our Discord and developer forums
-

Q12: What partnerships or integrations are planned?

A: We are actively pursuing:

- Academic partnerships with cryptography research groups
- Integration with existing DeFi protocols
- AI/ML platform collaborations
- Enterprise adoption pilots

Details will be announced as partnerships mature.

Chapter 15: Team & Recruitment

15.1 Core Talent Requirements

Cryptographers (PQC Experts)

- PhD in cryptography or equivalent experience
- Familiarity with NIST PQC standards
- Lattice-based cryptography research experience preferred

Blockchain Core Developers

- Rust development experience (Substrate/ink!)
- Distributed systems background
- Consensus algorithm understanding

AI/ML Engineers

- Machine learning model development
- ZK-SNARK/STARK proof systems
- AI Agent architecture design

Community & Developer Relations

- Blockchain community management experience
- Developer relations skills
- Multi-language communication capabilities

15.2 Join Us

Contact:

- Official Website: qylith.network
 - Developer Community: developers.qylith.network
 - Email: team@qylith.network
-

Chapter 16: References & Citations

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Closing: A Letter to Builders

Dear Builders,

We stand at a turning point in history.

Quantum computing is no longer science fiction—its footsteps are approaching. Existing internet infrastructure, including all major blockchains, will face unprecedented challenges in the quantum era. Simultaneously, AI agents are evolving from concepts into reality, and they will soon become residents of the digital world alongside humans.

The convergence of these two trends is both crisis and opportunity.

Qylith is our answer to the future. We are not patching the ruins of the old world—we are building skyscrapers on the foundations of a new one. We chose post-quantum cryptography from the genesis block, and we built a native environment for AI agents from the first line of code.

This is not an easy path. Implementing post-quantum cryptography is full of challenges, and an AI-native execution environment requires entirely new ways of thinking. But we believe this is the right path—the path that must be walked by someone.

If you believe:

- Cryptographic security is a non-negotiable底线
- AI agents will reshape the form of digital civilization
- Decentralized infrastructure is the guarantee of human freedom

Then, you are who we are looking for.

Whether you are a cryptographer, developer, designer, or simply someone passionate about this vision, we welcome you to join this construction.

The future is not waited for—it is built.

Let's start with Qylith and lay the foundation for the next era together.

Join Us:

- Website: qylith.network
 - Discord: discord.gg/qylith
 - 🐦 Twitter/X: [@QylithChain](https://twitter.com/QylithChain)
 - Contact: team@qylith.network
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The Qylith Team 2026

This document is the technical whitepaper v1.0 for the Qylith project. Specifications and timelines are current planning and may be subject to change. All technical parameters and roadmaps represent current plans and actual implementation may vary.

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